

Quantum Category $\mathcal{O}$, II

Affine Hecke Category

We have seen that the affine Weyl group controls certain aspects of $\mathcal{O}_\varepsilon^{\text{mix}}$. In fact, much more is true: there are derived equivalences between $\mathcal{O}_\varepsilon^\Theta$'s & certain “standard” categories associated to W^a : (singular) blocks of the affine Hecke category.

Definition

Let W' be a (possibly infinite) reflection group acting on its reflection representation V .

Example 1. $W' = W^a$, $V = \mathfrak{h} \oplus \mathbb{C}\hbar$, $W^a \curvearrowright V$: $\hbar$ is W^a -invariant, W acts on $\mathfrak{h}$ as usual, and for $\lambda \in \Lambda_0$, have $t_\lambda x = x + \langle \lambda, x \rangle \hbar$, $x \in \mathfrak{h}$.

Let $S \subset W'$ be the subset of simple reflections in W' . Set $R = S(V)$, this is a graded algebra.

Definition 1. Let $s \in S$. Define the Bott-Samelson R -bimodule $B_s = R \otimes_{R^s} R$ (where R^s is the subalgebra of s -invariants in R). Note that B_s is graded as a bimodule.

Consider the categories $R\text{-grbi} \bmod$ & $R\text{-gr} \bmod$ of graded finitely generated R -bimodules & graded R -modules. For $j \in \mathbb{Z}$, let $\langle j \rangle$ denote the endofunctor of grading shift:

$$B\langle j \rangle_i = B_{i+j} \quad (\text{for } B \in R\text{-gr}(\text{bi})\bmod)$$

We consider the following (additive) categories:

- **SBim** (the category of Soergel bimodules): the full subcategory in $R\text{-grbi} \bmod$ consisting of all graded bimodules obtained from R & B_s , $s \in S$, by $\otimes$, $\oplus$, $\langle j \rangle$, & taking direct summands.
- **SMod** (the category of Soergel modules), the full subcategory $\{B/BV \in R\text{-gr} \bmod \mid B \in \text{SBim}\}$. One can show it's closed under taking direct summands.
- **SMod** (the category of ungraded Soergel modules), a full subcategory in $R\text{-mod}$ whose objects are the same as in **SMod** (w. grading forgotten).

The categories **SBim**, **SMod** & **SMod** are "Krull-Schmidt": every object is isomorphic to a unique $\oplus$ of indecomposables. The Soergel categorification theorem describes the indecomposable objects in these categories, we will need a version for **SMod**.

Proposition 1. The indecomposable objects in **SMod** are in bijection w. W'

We are now in position to define the version of the affine Hecke category we need. Set $W' = W^a$ & $V = \mathfrak{h} \oplus \mathbb{C}\hbar$, see Example.

$\mathcal{H} := K^b(\underline{\text{SMod}})$, where K^b stands for the bounded homotopy category of complexes.

Derived Equivalence, Regular Blocks

Theorem 1 (I.L. 23). Suppose Θ is a free W^a -orbit (no stabilizer). Then $D^b(\mathcal{O}_\varepsilon^\Theta) \xrightarrow{\sim} \mathcal{H}$.

A somewhat weaker version of Thm was also obtained by R. Situ.

Remark 1. Such orbit exists $\Leftrightarrow d > h$ (h = Coxeter number of G , e.g. for $G = \text{SL}_n$, $h = n$). In this case $\Theta := W^a \cdot 0$ is free, the corresponding block is called principal.

Derived Equivalence, Singular Blocks

Consider the general (not necessarily free) Θ . We want a description of $\mathcal{O}_\varepsilon^\Theta$ similar to that in Theorem 1. First, we'll choose a distinguished element $\lambda^0 \in \Theta$. Set $\mathcal{A} = \{\lambda \in \Lambda_0 \mid \langle \lambda + \rho, \alpha_i^\vee \rangle \leq 0 \ \& \ \langle \lambda + \rho, \delta^\vee \rangle \geq -d\}$, where $\delta^\vee$ is the maximal coroot. Then $\mathcal{A}$ is a fundamental domain for $W^a \curvearrowright \Lambda_0$ (exercise). $\lambda^0 :=$ unique point in $\mathcal{A} \cap \Theta$.

For S take the set of reflections about the walls of $\mathcal{A}$. Set $J = \{s \in S \mid s \cdot \lambda^0 = \lambda^0\}$. $R^J = \bigcap_{s \in J} R^s$, polynomial algebra.

Definition 2. The category ${}_J \underline{\text{SMod}}$ is the full subcategory in R^J -mod whose objects are direct summands of objects in $\underline{\text{SMod}}$ (viewed as modules over $R^J \subset R$). Set $\mathcal{H} := K^b({}_J \underline{\text{SMod}})$.

Theorem 2 (I.L. 23). $D^b(\mathcal{O}_\varepsilon^\Theta) \xrightarrow{\sim} \mathcal{H}$.

Motivations

$\mathcal{O}_\varepsilon^{\text{mix}}$ was popularized by Gaitsgory who wanted an extension of the famous Kazhdan-Lusztig equivalence between

1. a certain parabolic category $\mathcal{O}$ for the affine Lie algebra $\widehat{\mathfrak{g}}$
2. and the finite dimensional representations of U_ε^l

He proposed to replace (i) w. the full cat. $\mathcal{O}$ & U_ε^l -mod_{fd} w. $\mathcal{O}_\varepsilon^{\text{mix}}$. His conjectures essentially yield our main equivalence & were proved by his students Chen & Fu by methods very different from mine (& with different assumptions on d). The motivation came from Quantum Geometric Langlands.

Another motivation: one should be able to use a connection between $\mathcal{O}_\varepsilon^{\text{mix}}$ and the affine Hecke category to get character formulas in the category of finite dimensional of U_ε^{DK} with some (currently a bit mysterious) additional “equivariance” structure, similar in spirit to Bezrukavnikov-I.L. 20.

Finally, a far reaching motivation is to understand the affine analog of $\mathcal{O}_\varepsilon^{\text{mix}}$: where instead of $U_\varepsilon^{\text{mix}}(\mathfrak{g})$ one has $U_\varepsilon^{\text{mix}}(\widehat{\mathfrak{g}})$ — understanding $\mathcal{O}_\varepsilon^{\text{mix}}$ is a necessary prerequisite.